



sTMS Interface Functional Specification

Nanyang Polytechnic – Billing Information

Version 1.0

Document Version

Version	Effective Date	Summary of Changes	Created/Updated By
0.1	25/01/2022	Initial version of NYP Billing Info IFS	AvePoint
0.2	08/03/2022	Revised based on feedback from NYP. 2.1 System Flow step2; Update Billing Info API: Add ReferenceDate in BillingQuery field; Add BillNo in Bill field; Add a new field named FeeInfo; Update parameter format for 3. Interface Fields and 6. Interface Schemas.	AvePoint
0.3	28/03/2022	Revised based on NYP feedback: Update 2.1 System Flow diagram; Indicate authentication method for 2.3 Interface Authentication; Add Annex A: Illustration of the computation of outstanding payments.	AvePoint
0.4	04/05/2022	Revised based on NYP feedback. 3.2 Interface Response Fields: Remove Financing Scheme section; Rename FinancingScheme in Receipt section to Payment Mode. - Add 5.2 Constraints section.	AvePoint
0.5	11/05/2022	Revised based on NYP feedback. 3.2. Interface Response Fields and 6. Interface Schemas: Add Charge Type, Course Code, Course Name for Bill, BillAdjustment and Receipt section. - Delete all comments, accept all changes and stop tracking.	AvePoint
0.6	18/05/2022	Revised based on NYP feedback. 3.1. Interface Launch Parameter and 6. Interface Schemas: Add course status as new launch parameter.	AvePoint
1.0	26/05/2022	Official version for acceptance and sign-off. Revised based on NYP feedback. 3.1. Interface Launch Parameter: Change the remarks for the CourseStatus field. - Delete all comments, accept all changes and stop tracking.	AvePoint



Document Acceptance Form

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Table of Contents

Document Version	1
Document Acceptance Form	2
1. Introduction	2
1.1 Purpose of this document	2
1.2 Intended Audience	2
2. Technical Overview	3
2.1 System Flow	3
2.2 Interface Design Summary	4
2.3 Interface Authentication	5
3. Interface Fields	6
3.1 Interface Launch Parameter	6
3.2 Interface Response Fields	6
4. Error Handling	9
4.1 Error Handling	9
5. Assumption and Constraints	10
5.1 Assumption	10
5.2 Constraints	10
6. Interface Schemas	11
7. Appendix	17

1. Introduction

1.1 Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirement and detailed technical specification between the SaaS shared Training Management System (sTMS) and Nanyang Polytechnic Common Application for student billing related information.

1.2 Intended Audience

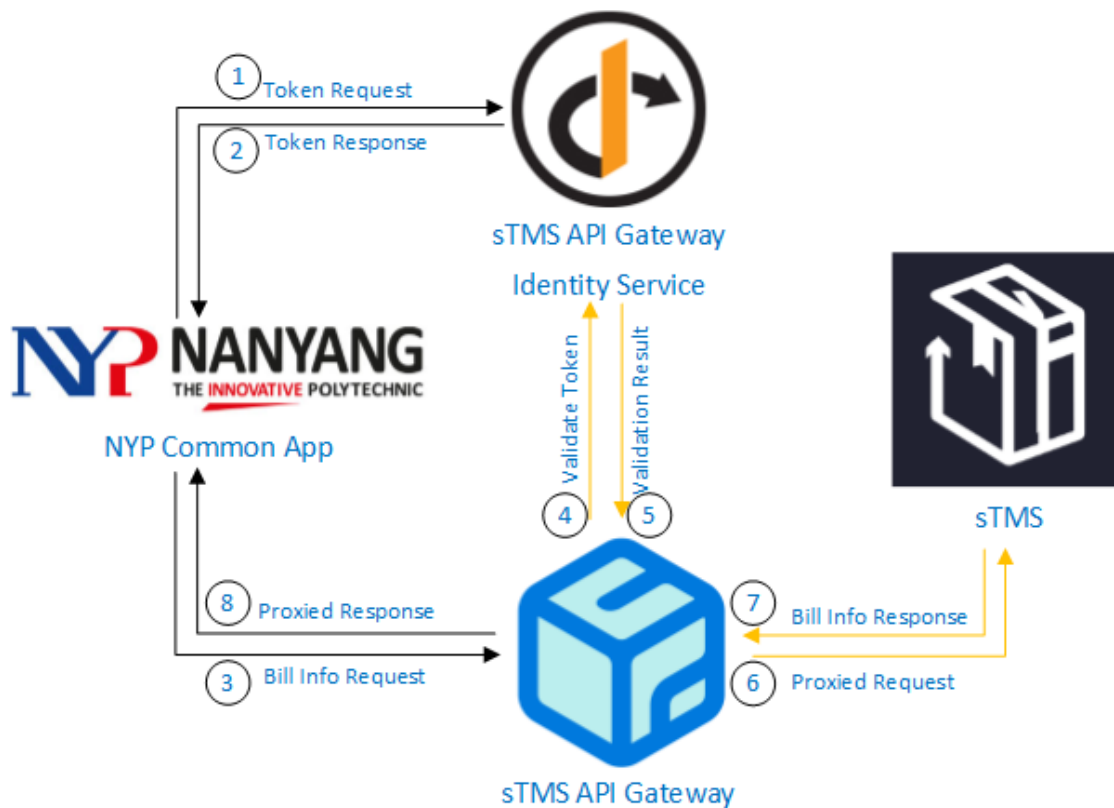
This technical document is used for system-to-system communication; therefore, the intended audiences are the technical stakeholders of both systems.

2. Technical Overview

There are 4 roles involved in the interface diagram.

- **NYP Common App:** client application that consumes the API.
- **sTMS API Gateway:** sTMS interface Integration middleware that validates HTTP request authentication, authorization, and payload inspection.
- **sTMS API Gateway Identity Service:** sTMS interface Identity Issuer that issues access token for API authentication and authorization
- **sTMS:** shared Training Management System.

2.1 System Flow



SN	Description
1	NYP Common App sends POST request with <i>client_id</i> and <i>secret</i> to sTMS API Gateway Identity Service to get API access token.

2	sTMS API Gateway Identity Service sends the access token back to NYP Common App within response message. This access token can be reused within 1 hour, client application can use an access token for multiple API request before the access token expires, after that client application needs to request a new access token again.
3	NYP Common App sends HTTP POST request with given student id, reference date and course status to sTMS API Gateway.
4	sTMS API Gateway sends request to sTMS API Gateway Identity Service to validate the access token of the billing info request.
5	sTMS API Gateway Identity Service sends back the validation result.
6	sTMS API Gateway verifies the request and sends a proxied request to sTMS.
7	sTMS queries the billing info by student id in database and sends response to API Gateway.
8	sTMS API Gateway sends the billing info response to NYP common app.

2.2 Interface Design Summary

Interface Name	GetBillingInfoNYP
Hosting System	shared Training Management System
Data Source System	shared Training Management System
Data Destination System	Nanyang Polytechnic Common Application
Mode of Interface	HTTP API
Method	POST
Content-Type	application/json
Authorization	Bearer <access_token>
Frequency	Realtime
Encryption Method	HTTPS
Endpoint URL	<TBC>

2.3 Interface Authentication

sTMS use OAuth (OAuth2.0) to perform authentication. NYP Common App needs to send HTTP POST request to sTMS Identity Service endpoint to get access token and put the access token into HTTP request 'Authorization' to consume target API.

sTMS Identity Service Endpoints:

Env	URL
UAT	<TBC>
Production	<TBC>

To access the API, the client application needs to authenticate using the client credential grant from API gateway for an access token with the following parameters:

- **grant_type**: REQUIRED. The value must be set to "client_credentials"
- **client_id**: REQUIRED. Client ID.
- **client_secret**: REQUIRED. Client Secret.

Sample:

```
curl --location --request POST '{URL}' \  
--header 'Content-Type: application/x-www-form-urlencoded' \  
--data-urlencode 'grant_type=client_credentials' \  
--data-urlencode 'client_id={client_id}' \  
--data-urlencode 'client_secret={client_secret}'
```

Response Body:

```
{  
  "access_token": "{access token}",  
  "expires_in": 3600,  
  "token_type": "Bearer",  
  "scope": "{api scope}"  
}
```


3. Interface Fields

3.1 Interface Launch Parameter

The asterisk * represents the field is required.

Name	Description	Data Type	Size	Remarks
StudentId*	Unique identifier of student	String	10	Student UAM common ID or NRIC/FIN
ReferenceDate	Reference date to sum up outstanding payments on and before the reference date	String(\$DD-MM-YYYY)	10	Format is DD-MM-YYYY ReferenceDate can be blank.
CourseStatus	Course status to filter all course-related bills or bills for courses the student has not graduated	String	10	Value list: Active, All, Blank. Active: All the bills of the courses taken by the student but have not graduated. All: All the bills of the courses taken by the student. Blank: All the bills (course related bills and non-course related bills) for the student.

3.2 Interface Response Fields

The asterisk * represents the field is required.

Name	Description	Data Type	Size	Remarks
FeeInfo				
TotalBalancePayable *	Sum of outstanding payments on and before the reference date	Number	10	Please see Annex A for the illustration.
Bill				
AcadYear*	Academic year in which the student is billed	Number	4	Sample values: 2021, 2022
Semester	Semester/Term in which the student is billed	String	2	S1, S2, Blank
BillNo*	Bill No.	String	20	NYPINV20220100000000 PREFIX defined by IHL + YYYYMM+8 digits of running number
BalancePayable *	Outstanding balance of the bill	Number	20	
PaymentDueDate*	Payment Due Date	String(\$DD-MM-YYYY)	10	Format is DD-MM-YYYY
Remarks	Remarks	String	255	

ChargeType*	Indicate bill charge type	String	40	Course, Q2 evaluation, Student service, Disciplinary fine, Supplementary fee
CourseCode	Course code	String	255	If have multiple course codes, value will be separated by . Sample value: NYP-SD-FQ0001 NYP-SD-FQ0002
CourseName	Course name	String	255	If have multiple course names, value will be separated by . Sample value: Specialist Diploma in Clinical Trials Management Specialist Diploma in Nutrition Science
BillAdjustment				
AcadYear*	Academic year in which the student is billed	Number	4	E.g. 2021, 2022
Semester	Semester/Term in which the student is billed	String	2	S1, S2, Blank
Description	Adjustment Description	String	255	
Type	Adjustment Type Currently, this is blank for CET.	String	10	Nil
BillNo*	Bill No.	String	20	NYPINV20220100000000 PREFIX defined by IHL + YYYYMM+8 digits of running number
SponsorCode	Sponsoring Company Name	String	100	
ChargeType*	Indicate bill charge type	String	40	Course, Q2 evaluation, Student service, Disciplinary fine, Supplementary fee
CourseCode	Course code	String	255	If have multiple course codes, value will be separated by . Sample value: NYP-SD-FQ0001 NYP-SD-FQ0002
CourseName	Course name	String	255	If have multiple course names, value will be separated by . Sample value: Specialist Diploma in Clinical Trials Management Specialist Diploma in Nutrition Science
Receipt				
AcadYear*	Academic year in which the student is billed	Number	4	Valid values: 2021, 2022
Semester	Semester/Term in which the student is billed	String	2	S1, S2, Blank
PaymentMode*	Payment mode of the receipt.	String	100	Value list: PSEA, SFC, Credit Card, eNETs, etc.
BillNo*	Bill No.	String	20	NYPINV20220100000000

				PREFIX defined by IHL + YYYYMM+8 digits of running number
ReceiptNo*	Receipt No	String	20	
ReceiptDate*	Receipt Date	String(\$DD-MM-YYYY)	10	Format is DD-MM-YYYY
BankReceiptDate	Bank Receipt Date Currently, this field is blank.	String	10	Nil
ReceiptRefNo	Reference of the receipt. Example, it can be the IBG reference No. or enets reference No.	String	20	
DepositAmount	Deposit Amount Currently, this is blank for CET students.	Number	10	Nil
ChargeType*	Indicate bill charge type	String	40	Course, Q2 evaluation, Student service, Disciplinary fine, Supplementary fee
CourseCode	Course code	String	255	If have multiple course codes, value will be separated by . Sample value: NYP-SD-FQ0001 NYP-SD-FQ0002
CourseName	Course name	String	255	If have multiple course names, value will be separated by . Sample value: Specialist Diploma in Clinical Trials Management Specialist Diploma in Nutrition Science
Award				
Title	Award Title	String	255	
Type	Award Type	String	20	Sample data: BURSARY
AwardYear	Year Awarded	Number	4	Sample data: 2020, 2021
Amount	Amount Awarded	Number	10	Sample data: 2000

4. Error Handling

4.1 Error Handling

The table below provides the detailed error listing and handling procedure.

Error Code	Description	Handling Procedure
403	The request is forbidden to access the specified resource.	Verify authorization header in your HTTP request; Raise a ticket to sTMS support to verify the application registration in API gateway.
404	The Student UAM common ID or NRIC/FIN cannot be found.	Verify the interface launch parameter in your HTTP request; Raise a ticket to sTMS support team for further investigation.
500	Internal server error	Raise a ticket to sTMS support for further investigation.

5. Assumption and Constraints

5.1 Assumption

No.	Assumptions
1	As sTMS will be the core system for all the transactions, therefore any unmapped billing information field from Nanyang Polytechnic Common Application will be kept empty.

5.2 Constraints

No.	Constraints
1	This interface contains all bills incurred to the student. ReferenceDate is only used to sum up the outstanding payments of this student, not as a filter condition for the bills.
2	Semester is not a mandatory field in sTMS (E.g., Short courses may not have semester info). If semester is not available for a course, Interface will leave the semester field value blank.

6. Interface Schemas

Interface Schemas are detailly described in the following table:

BillingQuery ▾ {		←
StudentId	string	
	Student UAM common ID.	
ReferenceDate	string(\$DD-MM-YYYY) example: 11-05-2022	
	Reference date to sum up outstanding payments on and before the reference date	
CourseStatus	string	
	Course status to filter all course-related bills or bills for courses the student has not graduated.	
}		

BillingInfo ▾ {		←
Fee	FeeInfo > {...} ←	
Bills	▾ [Bill > {...} ←]	
Adjustments	▾ [BillAdjustment > {...} ←]	
Receipts	▾ [Receipt > {...} ←]	
Awards	▾ [Award > {...} ←]	
}		

FeelInfo ▾ {

TotalBalancePayable

number

example: 6000

Sum of outstanding payments on and before the
reference date

}

Bill ▾ {

AcadYear*	number example: 2021, 2022 Academic year in which the student is billed
Semester	string example: S1, S2 Semester/Term in which the student is billed
BillNo*	string example: NYPINV20220100000000 PREFIX defined by IHL + YYYYMM+8 digits of running number
BalancePayable*	number example: 1000 Outstanding balance of the bill, unit is SG dollar
PaymentDueDate*	string(\$DD-MM-YYYY) example: 2021-12-31 Payment due Date
Remarks	string bill remarks
ChargeType*	string example: Course, Q2 evaluation Indicate bill charge type
CourseCode	string example: NYP-SD-FQ0001 NYP-SD-FQ0002 Course code
CourseName	string example: Specialist Diploma in Clinical Trials Management Specialist Diploma in Nutrition Science Course name

}

BillAdjustment ▾ {



AcadYear*	number	Academic year in which the student is billed
Semester	string <i>example: S1, S2</i>	Semester/Term in which the student is billed
Description	string	Bill adjustment description
Type	string	Currently, this is blank for CET
BillNo*	string <i>example: NYPINV20220100000000</i>	PREFIX defined by IHL + YYYYMM+8 digits of running number
SponsorCode	string	Sponsoring Company Name
ChargeType*	string <i>example: Course, Q2 evaluation</i>	Indicate bill charge type
CourseCode	string <i>example: NYP-SD-FQ0001 NYP-SD-FQ0002</i>	Course code
CourseName	string <i>example: Specialist Diploma in Clinical Trials Management Specialist Diploma in Nutrition Science</i>	Course name

}

Receipt ▾ {



AcadYear*	number	Academic year in which the student is billed
Semester	string example: S1, S2	Semester/Term in which the student is billed
PaymentMode*	string example: PSEA, SFC, Credit Card	Payment mode of the receipt.
BillNo*	string example: NYPINV20220100000000	PREFIX defined by IHL + YYYYMM+8 digits of running number
ReceiptNo*	string	Receipt No.
ReceiptDate*	string(\$DD-MM-YYYY) example: 11-05-2022	
BankReceiptDate	string(\$DD-MM-YYYY)	Bank Receipt Date. Currently, this field is blank
ReceiptRefNo	string	Reference of the receipt, it can be the IBG reference No. or enets reference No.
DepositAmount	number	Deposit Amount. Currently, this is blank for CET students
ChargeType*	string example: Course, Q2 evaluation	Indicate bill charge type
CourseCode	string example: NYP-SD-FQ0001 / NYP-SD-FQ0002	Course code
CourseName	string example: Specialist Diploma in Clinical Trials Management / Specialist Diploma in Nutrition Science	Course name

Award ▾ {

Title string

Award title

Type string

example: Bursary, Scholarship

Award Type

AwardYear number

Award year

Amount number

example: 2000

Year awarded

}



Avepoint_NYP_Billing
V1.0.0-resolved.json

7. Appendix

Annex A: Illustration of the computation of outstanding payments

Assume that Current Date is 23 Mar 2022. Billed Amount is \$100.

S/No	Payment Due Date	Reference Date	Date bill paid	Outstanding Payments returned by API
1	31 Jan 2022	Blank (which means reference date will be set as current date)	15 Feb 2022	0
2	31 Jan 2022	14 Feb 2022	15 Feb 2022	100
3	31 Jan 2022	15 Jan 2022	15 Feb 2022	100
4	20 Feb 2022	14 Feb 2022	15 Feb 2022	100
5	31 Mar 2022	Blank	Bill is not paid	100

Note: Outstanding payments are computed as Billed Amount - paid amount - deemed paid amounts where deemed paid amounts are payment made via PSEA, SkillsFuture Credits, etc.